Local Income Taxes and Residential Sorting
Pitch

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April 23, 2026

The question

- 17 states let local governments tax earned income[TF].
- Do people move to avoid income taxes or do they take on the cost?
- Can't find research specifically answering this question at the local level.
- Specifically looking at sub-county where standard migration data (ACS, IRS SOI) is too coarse. (most of studied state level policies)

Two margins — both need more research

- Rate changes (intensive): do higher rates push workers out? PA, MD [MD], MI, and KY [KY].
- Adoption (extensive): does turning a tax on cause sorting?:
  - School Districts
  - Cities
  - Counties
- Endogeneity of "why are you raising (or lowering?) income taxes?"

Why I want to start with Pennsylvania (intensive margin)

- Every city and school district levies an earned income tax which gives continuous variation on rates over geographies [rates].
- Every city files an Annual Financial Report so we can see what the money funds and what other taxes were raised or lowered [AFR]. (can directly address endogeneity concerns). Can easily make a scraper for this site.
- Act 32 (2012) centralized collection and made employer withholding binding a natural experiment for whether enforcement matters [Act 32].

Why LODES (origin–destination data)

- Links where you live to where you work, annually, at the Census block [LODES].
- I work with LODES data for my RA at Accessibility Observatory (familiarity)
- Use aggregate data at Census Tract to look at shifts within cities/school districts that the tract is in.
- When a city raises its rate, does the share of its workers who also live there fall?

How PA tax works

The worker pays the higher of the two rates, but how that bill is split between the cities is piecewise:

$$\left(\tau^{\text{home}},\ \tau^{\text{work city}}\right) = \begin{cases} \left(\tau^{\text{res}} - \tau^{\text{work}},\ \tau^{\text{work}}\right) & \text{if } \tau^{\text{res}} > \tau^{\text{work}} \\ \left(0,\ \tau^{\text{work}}\right) & \text{if } \tau^{\text{res}} \leq \tau^{\text{work}} \end{cases}$$

- Work city always collects its full nonresident rate.
- Home city only collects the leftover, and gets nothing if the work rate is higher.
- Which rate is higher also determines who can respond: home-binds $\rightarrow$ she moves her home; work-binds $\rightarrow$ she changes jobs.

Example: a $50,000 worker

- Case 1: lives in B (1.0%), works in A (0.5%). Bill = \$500. If B raises to 2.0%, bill = \$1,000. Home rate binds, so she escapes only by moving her home. Expect B's resident-worker count to fall.
- Case 2: lives in A (0.5%), works in B (1.0%). Bill = \$500. If B raises to 2.0%, bill = \$1,000. Work rate binds, so she escapes only by changing jobs. Expect B's total-jobs count to fall.
- LODES records each worker as a home-tract to work-tract flow, so either kind of move shows up directly in the data.

Methods (proposed)

Staggered event study on a muni × year panel:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell} \mu_\ell \cdot D_{it}^{\ell} + \varepsilon_{it}$$

- Outcomes of interest:
  - Share of a city's workers who also live there
  - Resident workers, nonresident workers, total jobs
  - Fiscal response (EIT revenue, spending by function)
  - Muni rate vs. school district rate
- Robustness: de Chaisemartin & D'Haultfoeuille (2024) continuous staggered estimator, which handles rate size rather than just on/off [DCDH 2024].